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Case Study 5

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Generative System Design Tool Digital Twin for Medical Devices

Background

Development and validation of a tool for generating a complete system design or module design for medical device including hardware (mechanical, electrical, electronic), embedded and application software and enclosure packaging.

A medical device system design typically begins from functional decomposition and mapping of existing devices/ modules/ systems to the individual functional and non-functional requirements.

AI/ Gen AI based tool potentially can leverage the data, datasheets and product information from open domain, the reference devices of similar functionalities and functional & non-functional requirements to 'generate' such system design with additional agents/ feedback mechanism for fine tuning of the parameters and creating the base system design. The candidate devices range form class-II devices used for invasive/ non-invasive health monitoring to the

Challenge

Develop innovative GenAI based solutions for generation of complete system design for select medical devices – one class-2, one class-3 (Example – Continuous Glucose Monitoring, Radiology/ CT MRI, Ablation Catheter, Ventilator, Dialysis Device etc).

- **Compliance**: Compliant to the medical devices system (excluding software) such as ISO 60601-1, ISO 62366-2, ISO 14971 (Risk Management).
- **Validation:** The design should be provable through simulated physical behaviour, electrical-electronic signals, interface protocols and IOs at critical checkpoints on temporal scale. The values should be logged/ viewable at any instance as snapshot of overall parameter status of the system (subsystems, components etc). Drill down and rollup view should be possible for the subsystems/ assemblies, components level and runtime value modification should be available for what-if analysis
- **Strategy:** Automation of the conventional techniques of system design and design time verification/ analysis resulting in reduced design time and avoidance of errors through what-if analysis.
- **Accuracy:** Generated architecture, design etc should be accurate in and be able to prove the different input-output level correctness with high fidelity of the model
- **Requirements:** The requirements will be provided as functional requirements with constraints such as usability, dimensions, applicable regulatory standards based on target regions. The testing should be possible through simulated functional scenarios as well as simulated behavior of components, subsystems, assemblies etc using what-if analysis. The 'fidelity' of the simulation should be adjustable from range of recorded data-based simulation to high-fidelity first principle-based simulations.

Evaluation Criteria

- **Effectiveness, Scenario Coverage:** The solution should be able to effectively cover all the functional and non-functional requirements (regulatory etc) and prove the effectiveness and fitness of the target system for the intended purpose. The models effectiveness should be proven quantitatively through parameters such as reduced size, reduced cost, complexity, ease of maintenance and obsolescence proofing etc.
- **Performance:** The systems created using the tool should prove to be comparably or better performant for existing and extendable functionalities of the future.
- **Scalability:** The solution created by the tool should be scalable so that the future growth of the input span of subsystems, complexity and functional requirements can be reasonably supported
- **Usability:** The solution should be easily usable for the system designers for inputting the requirements, constraints and what-if analysis.
- **Innovation:** The originality and creativity of the approach.

Potential Approaches

- **GenAI:** Use API-based or open-source GenAI tool and apply techniques such as prompt engineering/in-context learning/ RAG/fine tuning to reverse engineer/ understand the existing system designs, crawl through and collect data sheets/ manuals etc for available COTS information. GenAI techniques can be used for creating the new design at components, subsystems and assembly level including their interfaces and interactions. The first principle-based simulation can be mix of GenAI generated model and algorithmic models.
- **Stages:** Bottom-up approach can be used to start with component level model creation and its behavior + interface simulation. The same can be extended further including the interface/ integration/ interaction points identification and effect on other subsystems. Combining these can form overall system design-twin. Top-down approach can be explored and based on the evaluation, the final approach should be selected.
- **Working Implementation:** The implementation can preferably be based on open-source tools.
- **Performance comparison:**The accuracy and efficiency of multiple LLMs/ Gen Ai tools can be compared for the same solution. Suitable KPIs can be defined and employed to measure the performance.